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1. PROFESSIONAL SUMMARY & SYSTEMS PHILOSOPHY

Principal Systems Architect, Full-Stack Engineer, and Founding Technologist with 10+ years of end-to-end engineering experience spanning Google-scale web applications, distributed real-time media streaming platforms, physical clinic establishment, cross-platform mobile apps, and open-source developer tooling. Operates without artificial boundaries between software architecture, infrastructure, developer experience, and physical operations.

- **Google Site Kit Core Contributor (via 10up):** Partnered directly with Google engineering leadership as dedicated technical lead; authored approved design documents and merged **236 pull requests** directly into Google's upstream repository powering **3M+ production websites**.
- **Jasper Dual-Engine Architecture:** Architected a high-resilience media streaming daemon solving YouTube 403 Forbidden errors and stream dropouts via yt-dlp and FFmpeg, coupled with **Hosted Jasper**—a logically multi-tenant SaaS control plane for zero-token community onboarding.
- **Healthcare Startup Execution (Purrfect Universe):** Co-founded **Motion Mechanics Physiotherapy & Rehabilitation Hub** on CRP Road in Savar; directed commercial lease, physical clinic buildout, medical equipment procurement, and clinical team staffing, while engineering its complete paperless clinical operations platform (mm-website) with Next.js 15, Turborepo, tRPC, and PostgreSQL.
- **Global Open-Source Footprint:** Published and maintained React Native Paper libraries with **~7,000+ monthly downloads** on npm, official Arch Linux (AUR) packages, and Linux kernel hardware input interceptors for toddlers (jimha).

2. TECHNICAL ARSENAL & ARCHITECTURAL COMPETENCIES

Languages & Runtimes: TypeScript, JavaScript (ESNext, Node.js), Python, PHP, Rust, Shell/Bash, C/C++, SQL

Mobile & Cross-Platform: React Native, Expo, React Native Paper, React Navigation, Native Bridges

Backend & Distributed Systems: Node.js (Fastify, Express), Hexagonal Architecture (DomainPorts), Transactional Outbox Pattern

Databases & Persistence: PostgreSQL, MySQL, Prisma ORM, Redis (leases, pub/sub), Checksum-tracked Migrations, SQLite

Linux Systems & Fabrics: Arch Linux, Wayland, KDE Plasma 6, Virtual Displays, KRDP streaming, systemd, evdev, QEMU/KVM

Frontend & Web Platform: React (18/19), Next.js 15 (App Router, Server Components, SSR), Turborepo, Tailwind CSS

Type-Safe API Contracts: tRPC (end-to-end type safety), TanStack Query (React Query), TanStack Table & Virtual

Concurrency & Scale: Distributed Sharding, Concurrency Leases, Multi-Tenant Control Planes, Graceful Drains

Audio & Media Pipelines: Externalized Streaming Daemons, yt-dlp, FFmpeg Transcoding, PipeWire 8kHz PCM

Testing, Quality & CI/CD: Jest, React Testing Library, PHPUnit, Playwright, BackstopJS VRT, Docker, GitHub Actions, GCP

3. COMPREHENSIVE PROFESSIONAL EXPERIENCE

Principal Systems Architect, Co-Founder & Head of Tech/Operations

- **Jasper Media Platform & Hosted Jasper SaaS (sakibtamim/Jasper):** Architected dual-engine architecture for high-concurrency Discord audio streaming. Conquered YouTube anti-bot protections and 403 Forbidden errors by creating an externalized streaming daemon utilizing yt-dlp and FFmpeg. Invented "One Mind, Many Bodies" architecture with Automatic Feline Rotation (AFR), allowing dynamic token cycling across multiple bot workers.
- **Hosted Jasper SaaS Control Plane:** Engineered logically multi-tenant SaaS control plane for zero-token community onboarding. Implemented distributed worker sharding (HJ-OSS-11), per-guild concurrency leases (HJ-OSS-04), gateway intent isolation (HJ-OSS-03), interactive Next.js web client with synchronized audio seek bar (apps/web), typed Plugin SDK (HJ-OSS-10, HJ-OSS-12), and checksum-tracked PostgreSQL migrations (HJ-OSS-07).
- **Motion Mechanics Physiotherapy & Rehabilitation Hub (Purrfect Universe Planet):** Co-founded an outpatient physical rehabilitation clinic on CRP Road in Savar. Directed commercial lease negotiations, clinical space buildout, specialized medical hardware procurement, and clinical team staffing across Neurology, MSK, and Paediatrics.
- **mm-website Clinical Platform:** Architected proprietary clinic platform with Next.js 15, Turborepo, tRPC, Prisma ORM, PostgreSQL, TanStack Query, and Tailwind CSS. Built custom Google Cloud Platform re-auth wrappers and automated secrets deployment.
- **Sunshine Physio Universal App (PSL Client Project):** Delivered fully managed client system (sunshine-physio-universalapp): universal clinical triage and scheduling platform across Next.js and React Native / Expo with bilingual support (English/Bengali).
- **Pawthy Secrets CLI (@psl-oss/pawthy):** Authored and open-sourced zero-trust developer secret synchronization CLI adopted across internal repositories. Built with hexagonal architecture (DomainPorts), transactional outbox pattern, Fastify REST backend, and Discord approval workflows.
- **Knot Mesh & Linux Systems (kuasha420/knot-mesh):** Architected distributed multi-node Linux/Wayland workspace mesh connecting Arch Linux desktop, laptop, and Steam Deck via KRDP virtual monitor streaming, remote PTY execution, and Linda TupleSpace batching.
- **Technical Direction & Client Delivery Oversight:** Provided architectural leadership for Purrfect Download Manager (PDM concurrency design) and commercial client delivery for Karnaphuli Jewellery and Dream Emirates Suites.

10up

Remote (Global) | Aug 16, 2021 – Sep 11, 2024

Senior JavaScript Engineer (React / Full-Stack) — Dedicated Google Site Kit Lead

- **Google Site Kit Core Technical Lead (google/site-kit-wp):** Recruited directly on GitHub by 10up Head of Engineering. Partnered directly with Google engineering leadership on Google's official WordPress plugin active on **3M+ production websites worldwide**. Merged **236 production pull requests** across Google Analytics 4, Reader Revenue Manager, AdSense, and Search Console.
- **Site Kit: User Input v2 Design Document (Authored):** Approved by Felix Arntz (Google Tech Lead) and Mariya Moeva (Google Lead PM). Overhauled user goal-alignment questionnaire, local state persistence, asynchronous datastore syncing, inline editing, and personalized Key Metrics dashboard integration.
- **Site Kit: Ad Blocking Recovery (ABR) Design Document (Authored):** Approved by Mariya Moeva (Google Lead PM) and Evan Mattson (10up AD of Engineering). Architectural specification and rollout of AdSense ABR tags, automated snippet injection, publisher message creation, and BackstopJS Visual Regression Testing.
- **Existing Tags Simplification (ETS) Epic:** Unified tag detection and validation across GA4, Tag Manager, and AdSense using decoupled store factories (createExistingTagStore).
- **Process Innovation & Awards:** Created team-wide "Mid-Point Review" process officially adopted across 10up's Site Kit workflow to minimize PR churn. Awarded the **10up Summit "Uppie" Award** (Reykjavik, Iceland, 2023) for engineering excellence. Appointed orientation buddy/mentor for incoming Site Kit software engineers. Co-authored internal 10up TypeScript Training curriculum.

Star IT Limited & Aladdin Studios

Dhaka, Bangladesh | Feb 2020 – Aug 2021

JavaScript Engineer (React Native & Node.js) / Founder (Aladdin Studios)

- **28 Production Repositories Engineered:** Directed hands-on architecture across healthtech, matrimony, enterprise management systems, and media streaming platforms.
- **Priyojon Care Telemedicine Platform:** Engineered telemedicine ecosystem connecting patients with verified doctors for real-time video consultations, prescription delivery, and home healthcare provider dispatch across React Native mobile clients and Node.js REST APIs.
- **Farazi Homecare & Bismillah Matrimonial:** Built medical homecare mobile dispatch app and cross-platform matchmaking application with biometric authentication and profile discovery.
- **Enterprise Software & Media Streaming:** Engineered frontends and APIs for starnet (ISP subscriber management), irent (property accounting), ibos-erp, iBOSCRM, and ibotuber media streaming suite (authored 212 commits).

Junior Instructor, Department of Computer Technology

- Instructed polytechnic diploma students in C/C++, JavaScript, PHP, relational database design, and computer networking.
- Supervised practical laboratories on operating system administration, network routing protocols, and microcomputer hardware diagnostics.

Early Engineering & Freelance Systems

Savar, Dhaka, Bangladesh | 2012 – 2017

Software Developer & Linux Systems Administrator

- Published open-source WordPress plugins (country-state-dropdown-plugin in 2014) to WordPress Plugin Repository.
- Engineered raw PHP B2B licensing platforms (namm-b2b-license-delivery) with cryptographic key generation and validation.
- Built early hybrid mobile applications using Apache Cordova, PhoneGap, and Crosswalk Project (TUFFLA, PageDab). Managed bare-metal Linux servers and LAMP/LEMP stacks.

4. PUBLISHED OPEN-SOURCE LIBRARIES & COMMUNITY IMPACT

Library / Package	Registry	Monthly Usage	Technical Description
react-native-paper-phone-number-input	npm package	~4,100 / mo	High-performance international phone number input with search modal and flag picker for React Native Paper.
react-native-paper-toast	npm package	~2,140 / mo	Imperative toast notifications component, provider, and custom hook for React Native Paper.
react-native-immersive-bars	npm package	~570 / mo	Android navigation bar transparency component for immersive edge-to-edge layouts.
react-native-paper-alerts	npm package	~520 / mo	Cross-platform imperative alert and confirm modal dialogs for React Native (iOS, Android, Web).
mst-persistent-store	npm package	~240 / mo	Persistent MobX-State-Tree store provider and custom consumer hooks for React and React Native.
react-native-template-ts-plus	GitHub / npm	Production starter	Opinionated TypeScript starter template for React Native with navigation, state, and UI pre-wired.
antigravity-manager-bin	Arch Linux AUR	AUR package	Official Arch User Repository package build maintainer for Antigravity desktop tools.
jimha	GitHub (Python)	Linux evdev	Fullscreen tactile game intercepting raw hardware input via Linux evdev to protect host OS while entertaining toddlers.

5. EDUCATION & PROFESSIONAL CERTIFICATIONS**Diploma in Computer Technology (4-Year Engineering Diploma)**

Savar Digital Polytechnic Institute | Conferred 2017

Accredited technical diploma covering computer hardware architecture, electronics, operating systems, networking, and foundational software engineering.

B.Sc. in Computer Science & Engineering (for Diploma Holders)

European University of Bangladesh | 6 of 10 Semesters Completed

Completed core CS coursework (Data Structures, Algorithms, Advanced Programming). Stepped back to support spouse's CSE graduation through pregnancy and motherhood, and care for child JimHa.

Red Hat Certified Engineer (RHCE) & Cisco Certified Network Associate (CCNA)

IT Bangla Limited | Conferred 2017 & 2018

Enterprise Linux administration (systemd, security policies, storage, server hardening) and enterprise networking (routing, switching, TCP/IP stack).

6. HONORS, AWARDS & PERSONAL HORIZON

10up Summit Uppie Award (Reykjavik, Iceland, 2023): Awarded for outstanding engineering excellence and technical contributions on the Google Site Kit project.

Google Engineering Design Docs: Directly credited author of two canonical technical design documents approved by Google engineering leadership.

Feline Culture: Devoted pet parent to four cats; inspired the engineering naming of Purrfect Universe, Purrfect Software Limited, Jasper Music Platform, Purr Engine, Pawthy CLI, and Misty.

The Horizon (ChickenTech): Long-term trajectory to develop ChickenTech—a modern, free-range automated precision agriculture farm deploying edge sensors, microclimates, and solar monitoring.